


Uber Works by Uber

Uber is currently testing its new On-demand staff hiring business, Uber Works in Chicago. <https://digit.in/nov18-72-1a>


DJs can mix online streams

SoundCloud is rolling out a feature that allows DJ to perform by remixing songs streamed from its service. <https://digit.in/nov18-72-2>

Metal Whiskers

No, this isn't a story about Tony Stark's beard... or facial hair in '90s heavy metal bands...

Robert Sovereign-Smith |
editor@digit.in

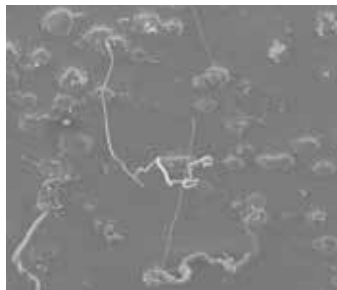
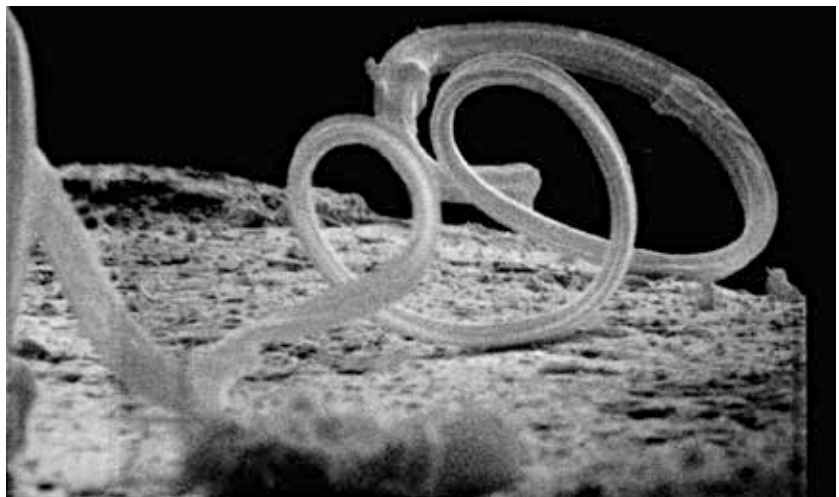
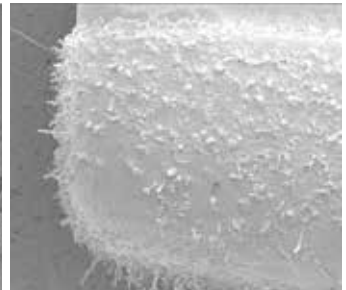
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or as long as we've been making circuits, we've had a problem with metals. While we think of metals as these smooth and cold materials with shiny polished surfaces, that's not exactly accurate most of the time. Add to that the race to miniaturise everything, and problems at the macro scale are several orders of magnitude worse at the micro or nano scale. This article is about one of those problems – namely, the weird tendency of some metals to grow “whiskers” spontaneously over time.

Yes, you read that right. Just as we humans grow body hair in all sorts of weird places, metals almost feel alive when viewed through a time lapse camera. Of course you'd have to lapse for years, and use microscopes, but if you did, you'd find that the smooth metal surfaces we normally envision just do not exist.

WHISKERING

So what is this whiskering we've mentioned a couple of times already? Well, just as men grow facial hair over a period of days or weeks, metals can grow little strands that look a lot like hair over months or years. We're not really sure of why it happens, or what causes



it, but scientists have made some educated guesses – none of which are conclusively proven. What we do know for sure is that it happens, and we have observed it happening. Now this might not seem like a problem for most applications, but when it comes to technology, it's a nightmare. This is because we have electricity flowing through our semiconductors, and connectors are all packed close together. Heck in the case of CPUs and GPUs, there are

hundreds of millions of transistors at a nanometer scale, all crowded into the same area as a pinhead!

With such precision engineering and such low fault tolerances, metal whiskering is a billion dollar pain in the backside for technology companies. Solder, for example, has been plagued with this problem for ages. This is because solder is usually tin or zinc based and the two are among the hairiest metals around.